

Name: Rushi Babasaheb Naikawade
 Branch: (AI&DS)
 Roll No: B-13
 Batch: SY-B1

AIM : Perform Data Ingestion and Cleaning: Acquire data and prepare it for analysis.

THEORY

1. What is data ingestion and why is it important in data science? Which Python library is commonly used for data ingestion and cleaning? Data ingestion is collecting data from various sources into a system for processing and analysis. It's vital in data science for providing highquality, structured data, forming the basis for all subsequent analysis and model building. pandas is commonly used for data ingestion and cleaning in Python.
2. Why is data cleaning necessary before performing analysis or machine learning? Data cleaning is essential for several reasons: Data cleaning is essential because raw data often contains errors, inconsistencies, or outliers. Clean data ensures accurate analysis, reliable insights, better model performance, and consistency across diverse data sources, ultimately leading to more efficient and trustworthy results.
3. How do you load a CSV file into a Pandas DataFrame? You can load a CSV file into a Pandas DataFrame using the `pd.read_csv()` function: `import pandas as pd df = pd.read_csv('your_file_name.csv')`
4. What is the difference between `head()` and `tail()` functions in Pandas? Both `head()` and `tail()` are DataFrame methods used to quickly inspect the rows of a DataFrame: `df.head(n=5)`: Returns the first n rows of the DataFrame. By default, it returns the first 5 rows. `df.tail(n=5)`: Returns the last n rows of the DataFrame. By default, it returns the last 5 rows.
5. How can you identify missing values in a dataset? You can identify missing values in a Pandas DataFrame using the `isnull()` (or `isna()`) method, often combined with `sum()` to get a count: `import pandas as pd Use df.isnull().sum()` to count missing values per column `df.isnull().sum().sum()` for the total missing values in the DataFrame. `df.isnull()` returns a boolean DataFrame indicating where values are missing. `import pandas as pd data = {'col1': [1, 2, None, 4], 'col2': ['A', 'B', 'C', None]} df = pd.DataFrame(data) print(df.isnull().sum())`

```
import pandas as pd
import numpy as np

data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'Alice', np.nan],
    'Age': [25, 30, 200, 25, 22],
    'City': ['New York', 'Los Angeles', np.nan, 'New York', 'Chicago'],
    'Sales': ['100', '200', '150', '100', '300']
}
df = pd.DataFrame(data)
# Explicitly cast 'Age' to float to handle potential median calculations later
df['Age'] = df['Age'].astype(float)
print("---- Original Data ----")
print(df)

df = df.drop_duplicates()
df['City'] = df['City'].fillna('Unknown')
df = df.dropna(subset=['Name'])
df['Sales'] = df['Sales'].astype(int)

median_age = df[df['Age'] < 100]['Age'].median()
df.loc[df['Age'] > 100, 'Age'] = median_age

print("\n--- Cleaned Data ---")
print(df)
print("\n--- Data Info ---")
df.info()
```

```
--- Original Data ---
   Name  Age  City Sales
0  Alice   25  New York  100
1   Bob   30  Los Angeles  200
2  Charlie 200      NaN   150
3   Alice   25  New York  100
4    NaN   22   Chicago  300

--- Cleaned Data ---
   Name  Age  City Sales
0  Alice 25.0  New York  100
1   Bob 30.0  Los Angeles  200
2  Charlie 27.5   Unknown  150

--- Data Info ---
<class 'pandas.core.frame.DataFrame'>
Index: 3 entries, 0 to 2
Data columns (total 4 columns):
```

```
#   Column  Non-Null Count  Dtype
---  -
0   Name    3 non-null      object
1   Age      3 non-null      float64
2   City     3 non-null      object
3   Sales    3 non-null      int64
dtypes: float64(1), int64(1), object(2)
memory usage: 120.0+ bytes
/tmp/ipykernel_16199/1457510752.py:20: FutureWarning: Setting an item of incompatible dtype is deprecated and will raise an
df.loc[df['Age'] > 100, 'Age'] = median_age
```

Conclusion : Data ingestion and data cleaning are essential steps in the data analysis process because they ensure that raw data is collected, organized and prepared properly before further processing. Data ingestion helps gather data from different sources into a structured format, while data cleaning improves data quality by removing errors, duplicates and missing values. Using tools like the Pandas library makes it easier to load datasets, preview them using functions like `head()` and `tail()`, and identify missing values effectively.